The Geometry of What Returns: From Quarks to Memory Wells via a Single Collapse Axiom

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A single axiom — *"Collapse is generative; only what returns is real"* — and the budget cost function $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$ of the Universal Measurement Contract Protocol (UMCP) reveal a recurring structural pattern across twenty orders of magnitude in scale: structure accumulates through iterated collapse-return cycles, and what survives each cycle is measurable by the same diagnostic kernel. We trace the geometry of what returns from quarks (10^{-15} m) through nuclei, atoms, stars, galaxies, neurons, and memory wells, encoding 37 entities across 8 categories as 8-channel trace vectors — with channel values assigned from domain-specific observables — and computing the kernel invariants $(F, \omega, S, C, \kappa, \mathrm{IC})$ at every level.

At each phase boundary in this scale ladder, the same two-step pattern operates: *geometric slaughter* — a single dead channel in the trace vector destroys the multiplicative coherence (IC) while the arithmetic mean (F) survives — followed by recovery through new degrees of freedom. Quarks lose coherence at the confinement boundary (IC/F: $0.957 \to 0.718$); nuclear binding restores it (IC/$F \to 0.954$). The same pattern reappears in biology: PTSD fragments four of eight channels, dropping IC/F from 0.982 (healthy memory) to 0.760 — the same diagnostic signature as the confinement cliff. The accumulation pattern is triple: space accumulates as iterated $|\kappa|$ (mass is accumulated well depth), time emerges from the rate and cost of return cycles (c_1/Γ'), and memory accumulates as well geometry shaped by repeated traversal. Ten Tier-2 theorems, verified by 129 automated tests (0 failures), formalize these three modes of accumulation and the cross-scale pattern they produce.

I. INTRODUCTION

What does a quark have in common with a traumatic memory?

The answer is precise: both exhibit IC/$F \approx 0.7$, both have dead trajectory-closure channels ($c_5 \leq 0.15$), both are classified as Collapse with thin-arc lensing morphology, and both occupy structurally identical positions on the budget surface $\Gamma(\omega)$. This is not an analogy. It is a diagnostic measurement — one whose result depends on how the systems are encoded, but whose pattern, once the encoding is fixed, is structural and reproducible.

The question sounds absurd until one examines the mathematics. The Universal Measurement Contract Protocol (UMCP) [1, 2] is a contract-first validation framework built on a single axiom: *"Collapse is generative; only what returns is real"* (Axiom-0). Its six-output kernel maps coherence coordinates $\mathbf{c} \in [0,1]^n$ to invariants $(F, \omega, S, C, \kappa, \mathrm{IC})$ via three provably exact identities [3]:

$$F + \omega = 1, \quad \mathrm{IC} \leq F, \quad \mathrm{IC} = e^{\kappa}. \tag{1}$$

The kernel is domain-agnostic by construction. It does not know whether its inputs describe a proton's internal structure or a patient's psychological state. Yet the same budget cost function

$$\Gamma(\omega) = \frac{\omega^p}{1-\omega+\varepsilon}, \quad p = 3,\ \varepsilon = 10^{-8}, \tag{2}$$

produces structural signatures that recur across scales with the same diagnostic values.

This paper traces those signatures from the smallest known structures to the largest, and from there into the living body and the human mind. The narrative is organized as a scale ladder, and at every rung the same three questions are asked:

1. What is the fidelity F (arithmetic mean of channels)?
2. What is the integrity IC (geometric mean, via $\mathrm{IC} = e^{\kappa}$)?
3. What is the heterogeneity gap $\Delta = F - \mathrm{IC}$, and what does it reveal?

The answer, across 37 entities spanning 10^{-15} m to macroscopic scales, is always the same: the gap Δ measures channel heterogeneity. Large Δ means one or more channels are near zero while others remain high — a signature we call *geometric slaughter*, because the geometric mean is destroyed by its weakest factor while the arithmetic mean barely notices. This mechanism operates with the same mathematics at the confinement boundary (quarks $\to$ hadrons) and at the trauma boundary (healthy recall $\to$ PTSD).

The claim is not that quarks and memories are "the same." It is that a domain-agnostic diagnostic lens, when applied to systems encoded along the same eight dimensions, reveals a recurring accumulation pattern: structure builds through iterated collapse-return, and the diagnostic invariants (F, IC, Δ) measure how that accumulation

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proceeds at each scale. The pattern depends on the encoding — the channel values are assigned from domain-specific observables, not measured by the kernel itself — but given a consistent encoding, the pattern is structural and reproducible.

A note on how this paper is built. The kernel is domain-agnostic: it receives any trace vector and produces the same structural diagnostics. This paper is structured analogously. Section II defines the fixed diagnostic toolkit and concludes with a compact *Diagnostic Key* (Sec. II A) — a reference card sufficient to enter any scale section independently. Sections III–VII are independent demonstrations that the same pattern — geometric slaughter at phase boundaries, recovery through new degrees of freedom — recurs at every scale. A reader may begin at the section matching their expertise: a particle physicist at Section III, a clinician at Section VII, an astrophysicist at Section V. Each section makes its case given the Diagnostic Key; cross-scale comparisons reference the Key's anchors (item 7) rather than requiring prior sections.

Linear reading enriches the pattern without being prerequisite. Read sequentially, the paper compounds: Section III gives us the gap Δ; Section IV shows the gap closes at phase boundaries; Section V reveals the gap as a geometric classifier (lensing morphology); Section VI shows the pattern is predictive; Section VII derives clinical consequences from the accumulated geometry. Each section retroactively deepens every previous one. But this compounding is an enrichment, not a gate: any single section, read with the Diagnostic Key, makes its case independently. A methodological caution: a reader who enters at a late section should verify that the pattern holds independently at other scales, and a linear reader should periodically check whether the accumulation of pattern-recognition is driven by the data or by the narrative momentum.

Five claims — three structural, two mathematical. This paper makes five claims about the accumulation pattern. Each is verified computationally by the theorem suite (Sec. IX). The first three are *structural*: they describe patterns that emerge from the encoding and are falsifiable *given a fixed encoding scheme* (Remark 2). The last two are *mathematical*: they follow from the kernel's definitions and hold for any valid input.

1. **Space accumulates as iterated** $|\kappa|$ (Theorem 9): Well depth grows linearly with iteration count: $\mathrm{depth}(N) = N\,|\Delta\kappa|$. Mass is accumulated well depth from repeated collapse-return cycles. *Falsified by*: an entity, under the same encoding, whose well depth does not grow linearly with iteration count.

2. **Time emerges from return cost** (Definition 4, Theorem 10): Effective temporal resolution $\mathcal{T} = c_1/\Gamma'$ orders all 37 entities consistently, from the photon ($\mathcal{T} \approx 0$) to the neutron star ($\mathcal{T} \approx 18$), with cost asymmetry producing the arrow of time (Theorem 11). *Falsified by*: an entity whose independently established temporal ordering contradicts the $\mathcal{T}$ ranking. Note that $\mathcal{T}$ is computed from the encoding, so this test requires an external temporal benchmark not derived from the channel values.

3. **Memory accumulates as well geometry** (Definition 3, Theorems 9, 14): The confinement cliff (IC/F: $0.957 \to 0.718$) and the trauma cliff (IC/F: $0.982 \to 0.760$) share the same dead channels (c_5, c_7) and the same mechanism. Recovery restores dead channels at each phase boundary. *Falsified by*: a phase boundary, under the same encoding, where the IC/F drop is not explained by dead channels.

4. **Budget gradient structures the manifold** (Theorem 5): $\mathrm{d}\Gamma/\mathrm{d}\omega > 0$ everywhere — the budget surface is monotonic, always attractive, and determines the cost landscape through which all systems evolve. This is a mathematical property of $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$ with $p = 3$; it holds independently of the encoding.

5. **Structural consistency across all scales** (Theorem 14): All 37 entities in 8 categories obey the three kernel identities ($F + \omega = 1$, $\mathrm{IC} \le F$, $\mathrm{IC} = e^{\kappa}$). These identities are mathematical theorems that hold for *any* input in $[0,1]^n$ — they confirm that the implementation is correct, not that the encoding captures something physical.

II. THE KERNEL

Definition 1 (GCD Kernel)**.** *Given coordinates* $\mathbf{c} = (c_1, \ldots, c_n) \in [\varepsilon, 1-\varepsilon]^n$ *and weights* $\mathbf{w} = (w_1, \ldots, w_n)$ *with* $\sum_i w_i = 1$*, the kernel computes:*

$$F = \textstyle\sum_i w_i\, c_i, \tag{3}$$
$$\omega = 1 - F, \tag{4}$$
$$S = -\textstyle\sum_i w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big], \tag{5}$$
$$C = \mathrm{std}(\mathbf{c})\,/\,0.5, \tag{6}$$
$$\kappa = \textstyle\sum_i w_i \ln(c_i + \varepsilon), \tag{7}$$
$$\mathrm{IC} = \exp(\kappa). \tag{8}$$

The heterogeneity gap $\Delta \equiv F - \mathrm{IC} \ge 0$ measures channel heterogeneity: $\Delta = 0$ iff all c_i are equal.

Lemma 2 (Heterogeneity Gap Non-Negativity)**.** $\Delta = F - \mathrm{IC} \ge 0$ *for all* $\mathbf{c} \in [\varepsilon, 1-\varepsilon]^n$ *and all weight vectors* $\mathbf{w} \in \Delta^n$*, with equality iff all* c_i *are identical.*

Proof. By Jensen's inequality applied to the concave function ln: $\sum w_i \ln c_i \le \ln(\sum w_i c_i)$. Exponentiating both sides gives $\mathrm{IC} = \exp(\sum w_i \ln c_i) \le \sum w_i c_i = F$. Equality holds iff all c_i are equal, since ln is strictly concave. Computationally verified to $< 10^{-12}$ residual across all 37 entities (Theorem 14). □

Definition 3 (Geometric Slaughter)**.** *A trace vector* $\mathbf{c}$ *exhibits* geometric slaughter *when at least one channel* $c_j \ll 1$ *while* F *remains moderate. Because* $\mathrm{IC} = \exp(\sum w_i \ln c_i)$ *and* $\ln(c_j) \to -\infty$ *as* $c_j \to 0$*, a single near-zero channel forces* $\mathrm{IC} \to 0$ *regardless of the remaining channels. Formally: for* $n = 8$ *equal-weight channels with one channel at* $c_j = \varepsilon$ *and the rest at* $c_k = 0.9$*,* $F = 0.787$ *but* $\mathrm{IC} = 0.079$ *— a* $10\times$ *collapse in multiplicative coherence from a single dead channel.*

Definition 4 (Effective Temporal Resolution)**.** *The* effective temporal resolution *of a system is* $\mathcal{T} \equiv c_1/\Gamma'(\omega)$*, where* c_1 *is the cycle return rate (fraction of collapse-return cycles that complete) and* $\Gamma'(\omega) = \mathrm{d}\Gamma/\mathrm{d}\omega$ *is the budget cost per cycle. The ratio is natural: each completed cycle (*c_1*) is one temporal "tick," and each tick costs* Γ' *units of budget. A system's internal time runs at the rate at which it can afford to tick — the number of ticks available per unit of budget. High* $\mathcal{T}$ *means many cheap ticks (fast internal time); low* $\mathcal{T}$ *means few or expensive ticks (slow or frozen time). This is the temporal mode of the accumulation pattern: time does not flow uniformly through the system; it accumulates at a rate set by the cost of return.*

Remark 1 (Why c_1/Γ' and Not Another Combination)**.** *The ratio* c_1/Γ' *is the simplest combination of completed cycles (*c_1*, dimensionless rate) and budget cost per unit drift (*Γ'*, dimensionless cost) with the operational meaning "ticks per unit cost." Alternative combinations produce orderings that contradict domain knowledge: the product* $c_1 \cdot \Gamma'$ *would make the black hole (*$c_1 = 0.50$*,* $\Gamma' = 0.48$*) temporally* faster *than the neutron star (*$c_1 = 0.98$*,* $\Gamma' = 0.054$*), because the black hole's higher cost would count as higher temporal resolution. The difference* $c_1 - \Gamma'$ *lacks scale invariance: adding a constant to all costs would change the ordering. Nevertheless,* $\mathcal{T}$ *is a* diagnostic definition*, not a derivation from first principles — it was selected because it produces physically sensible orderings across all 37 entities, and this circularity should be noted: the definition is partly validated by the ordering it produces. An independent derivation of temporal resolution from the kernel axioms alone remains an open problem.*

The budget surface

$$z(\omega, C) = \Gamma(\omega) + \alpha\, C, \quad \alpha = 1.0, \tag{9}$$

augments the kernel with the cost function (2). All frozen parameters ($p = 3$, $\varepsilon = 10^{-8}$, $\alpha = 1.0$, $\mathrm{tol}_{\mathrm{seam}} = 0.005$) are seam-derived, not prescribed [4].

The trace vector $\mathbf{c}$ has 8 channels, listed in Table I, with equal weights $w_i = 1/8$. These channels are abstract enough to encode any system that cycles through collapse and return: a quark bound inside a proton, a neutron star pulsing X-rays, a neuron firing action potentials, or a mind replaying a memory.

TABLE I. Eight-channel encoding. All values in $[\varepsilon, 1-\varepsilon]$.

i	Channel	Physical Meaning
0	coherence_persist.	Fraction of κ retained per cycle
1	cycle_return_rate	Fraction of cycles with $\tau_R < \infty$
2	well_depth_norm	Accumulated $\lvert\kappa\rvert$ (normalized)
3	gradient_strength	$\mathrm{d}\Gamma/\mathrm{d}\omega$ (normalized)
4	tidal_symmetry	$1 - \lvert\text{asymmetry}\rvert$ of $\mathrm{d}^2\Gamma$
5	trajectory_closure	Loop closure gap in (F, κ, C)
6	circulation_area	Enclosed area of collapse-return loop
7	heterogeneity_prof.	$1 - \Delta/F$ (channel uniformity)

Remark 2 (Encoding Methodology)**.** *Channel values are assigned from domain-specific observables, not measured by the kernel itself. The quark's trajectory closure* $c_5 = 0.85$ *reflects QCD confinement physics; the PTSD well's* $c_5 = 0.10$ *reflects clinical assessment of narrative fragmentation. The kernel does not discover these values — it receives them and computes the structural diagnostics (*F*,* IC*,* Δ*) that follow. What the kernel* does *reveal is the* pattern*: given a consistent encoding, the same diagnostic signatures (geometric slaughter, recovery through new degrees of freedom, cost-modulated temporal resolution) recur at every scale. The structural parallels are consequences of the encoding in combination with the kernel's mathematics — the question this paper investigates is whether the recurrence of these patterns across twenty orders of magnitude reflects a genuine structural motif or an artifact of the encoding scheme. Section VIII returns to this question.*

A. Diagnostic Key

The following reference card is sufficient to enter any scale section of this paper independently. Each section applies these diagnostics to a different domain; the structural finding is that the same pattern emerges regardless of entry point.

1. **Three identities** (exact for all inputs): $F + \omega = 1$, $\mathrm{IC} \le F$, $\mathrm{IC} = e^{\kappa}$.
2. **Three questions** at every scale: What is F? What is IC? What is $\Delta = F - \mathrm{IC}$?
3. **Geometric slaughter** (Def. 3): one near-zero channel kills IC while F survives.
4. **Two-step pattern**: slaughter at phase boundaries $\to$ recovery through new degrees of freedom.

TABLE II. Subatomic entities.

Entity	F	IC	Δ	IC/F
Up quark	0.766	0.734	0.033	0.957
Electron	0.656	0.515	0.141	0.784
Proton	0.476	0.342	0.134	0.718
Neutron	0.385	0.290	0.095	0.753
Photon	0.263	0.018	0.245	0.067

5. **Temporal resolution** (Def. 4): $\mathcal{T} = c_1/\Gamma'(\omega)$ — ticks per unit cost.

6. **Budget surface**: $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$; gradient positive, convex, pole at $\omega \to 1$.

7. **Reference anchors** (for cross-scale comparison): neutron star IC/F = 0.998, dust lane IC/F = 0.773, free quark IC/F = 0.957, proton IC/F = 0.718, photon IC/F = 0.067.

8. **Lensing classification**: perfect ring ($\Delta < 0.02$), thick arc ($0.02 \leq \Delta < 0.10$), thin arc ($0.10 \leq \Delta < 0.30$), distorted ($\Delta \geq 0.30$). No entity in this catalog reaches the distorted threshold. (Formalized as Thm. 12 in Sec. V.)

III. SCALE I: THE SUBATOMIC WORLD

Five subatomic entities — the up quark, the electron, the proton, the neutron, and the photon — are encoded as 8-channel trace vectors and passed through the kernel. The results (Table II) reveal the first structural drama of the scale ladder.

The quark is highly coherent. Within a hadron, the up quark has IC/F = 0.957 — nearly all channels are active, the geometric mean closely tracks the arithmetic mean, and the heterogeneity gap is a slim $\Delta = 0.033$. Its coherence persistence ($c_0 = 0.90$), trajectory closure ($c_5 = 0.85$), and heterogeneity profile ($c_7 = 0.80$) are all healthy. No channel is dead; no channel drags the geometric mean down. This number — 0.957 — will become the benchmark against which every subsequent cliff is measured, and the three channels just named are the ones that confinement will destroy.

The electron: a lepton that does not confine. The electron sits between the quark and the proton at IC/F = 0.784. Unlike quarks, electrons exist freely — they are not confined inside composites. Their gap $\Delta = 0.141$ comes not from dead channels but from the moderate values across several channels: the electron's well is shallow (low c_2) and its gradient modest (low c_3), reflecting its light mass and diffuse wavefunction. The heterogeneity is intrinsic, not imposed by confinement. This distinction matters: the electron's IC/F will reappear in the rocky planet (0.875) and the cortical network (0.974) — systems where coherence arises from lightness and flexibility rather than from deep binding.

The confinement cliff. Quarks cannot exist freely; they are confined inside hadrons. This confinement is visible in the kernel as a dramatic drop in IC/F:

$$\left.\frac{\mathrm{IC}}{F}\right|_{\text{quark}} = 0.957 \quad \longrightarrow \quad \left.\frac{\mathrm{IC}}{F}\right|_{\text{proton}} = 0.718. \tag{10}$$

The proton has $F = 0.476$ — moderate arithmetic coherence. But its trajectory closure ($c_5 = 0.15$) and heterogeneity profile ($c_7 = 0.30$) are near-dead: the confined quarks cannot close trajectories at the composite scale, and the internal structure is violently heterogeneous. These are precisely the channels that were healthy in the free quark ($c_5 = 0.85$, $c_7 = 0.80$) — confinement does not lower all channels uniformly; it selectively kills the ones that require freedom of motion. The dead channels destroy the geometric mean through geometric slaughter. The gap $\Delta = 0.134$ is four times the quark's, and the lensing classification has shifted from *thick arc* to *thin arc*. This cliff is not unique to the subatomic world. Binding restores exactly these two channels at the nuclear scale. At the cognitive scale, avoidance behavior and narrative fragmentation kill exactly these two channels again.

The photon: maximum slaughter. The photon achieves IC/F = 0.067 — the lowest ratio in the entire 37-entity catalog. Four of its eight channels are near-zero (well depth, gradient, trajectory closure, circulation area), reflecting its massless, non-orbiting nature. Its $F = 0.263$ is sustained by high coherence persistence ($c_0 = 0.99$) and tidal symmetry ($c_4 = 0.99$), but the geometric mean is annihilated. The photon is the pure case of geometric slaughter: half the channels are perfect, half are dead, and the dead ones win. Among the dead channels, trajectory closure ($c_5 \approx 0$) is the most decisive — the photon travels in straight lines through flat spacetime, closing no orbits. At the cosmic scale, the same structural signature appears in the black hole ($c_5 = 0.10$, Diagnostic Key anchor), where trajectories cannot close because nothing returns from beyond the horizon. The photon and the black hole are structural kin: both suffer from the impossibility of closed return, and both are classified as *thin arc* lensing. When light bends around a black hole to form an Einstein ring, the lensing agent and the lensed signal share the same dead channel.

These three numbers — IC/F = 0.957 for the quark, IC/F = 0.718 for the proton, IC/F = 0.067 for the photon — are the signature of geometric slaughter.

What we know so far — and what we do not. We have learned exactly one thing: the geometric mean is destroyed by its weakest factor while the arithmetic mean barely notices. At this point, Δ is just a number — the distance between two averages. It does not yet tell us about geometry, or gravity, or time, or clinical diagnosis. But notice one channel we have not yet discussed: c_1, the cycle return rate. The quark returns at $c_1 = 0.98$ — nearly every collapse cycle completes. The

TABLE III. Nuclear and atomic entities.

Entity	F	IC	Δ	IC/F
Helium-4 nucleus	0.759	0.695	0.064	0.916
Iron-56 nucleus	0.770	0.734	0.036	0.954
Carbon-12 atom	0.713	0.640	0.073	0.898

proton matches it ($c_1 = 0.99$): confinement kills trajectories but not the return rate. The photon, however, has $c_1 = 0.01$ — it almost never returns to its prior state, because it propagates at c and does not orbit. The neutron is intermediate ($c_1 = 0.60$): its β-decay lifetime is finite, so some cycles do not complete. At the subatomic scale alone, c_1 is just a number. But it is the seed of temporal resolution: when combined with the budget function $\Gamma'(\omega)$ (Diagnostic Key, item 5), the return rate becomes a clock and the gap becomes a lens.

IV. SCALE II: NUCLEAR RECOVERY

When quarks are confined inside hadrons, their trajectory-closure and heterogeneity channels die, dropping IC/F from 0.957 (free quark) to 0.718 (proton) — the *confinement cliff* of Table II. But confinement is not the end. When protons and neutrons bind into nuclei, new degrees of freedom emerge — magic numbers, pairing correlations, shell structure — and coherence recovers.

The helium-4 nucleus (the alpha particle) is doubly magic: closed proton and neutron shells produce IC/F = 0.916, recovering from the proton's 0.718. The recovery is not uniform — it is targeted. The proton's dead trajectory closure ($c_5 = 0.15$) climbs to $c_5 \approx 0.80$ in the alpha particle, because nuclear binding provides a closed orbital within which the previously confined degrees of freedom can circulate. The heterogeneity profile (c_7) similarly recovers: four nucleons in a symmetric shell produce far more uniform channel distributions than the three quarks locked inside a single hadron. The channels that confinement killed are precisely the channels that nuclear binding restores.

Iron-56 sits at the peak of the nuclear binding energy curve and achieves IC/F = 0.954 — nearly as coherent as the free quark. Peak binding energy per nucleon corresponds to peak IC/F: the energetic optimum and the coherence optimum coincide because binding energy measures coupling depth while IC/F measures how uniformly that coupling distributes across channels. Iron-56 is also the end product of stellar fusion — the element at which a star's collapse-return engine reaches thermodynamic equilibrium.

Carbon-12, the basis of organic chemistry, holds IC/F = 0.898. Its three alpha clusters produce a moderately homogeneous structure: high enough to sustain complex bonding, heterogeneous enough to permit the chemical flexibility that organic molecules require. At the biological scale, the question of why life is carbon-based has a structural answer: carbon sits exactly where coherence is high enough for stability but the gap $\Delta = 0.073$ is wide enough for combinatorial diversity. It is neither iron (too tightly bound) nor helium (too symmetric to branch).

The mechanism is clear: nuclear binding creates new channels of coherence (shell filling, pairing, trajectory closure at the nuclear scale) that replace the dead channels of the confined hadrons. *Recovery happens when new degrees of freedom replace dead ones at a phase boundary.* The proton's c_5 was dead because quarks cannot close orbits inside a hadron; the alpha particle's c_5 is alive because nucleons *can* close orbits inside a nucleus. The replacement is specific, not generic: the new structure does not raise all channels — it targets the dead ones.

This principle — collapse destroys coherence; new structure restores it — is the central motif of the scale ladder. It operates identically at the transition from diffuse matter to stars, and at the transition from neural substrate to organized memory.

A deeper implication. Two facts are now in evidence: (i) dead channels kill IC (geometric slaughter), and (ii) new channels restore it (recovery at phase boundaries). But these two facts together imply something neither contains alone: *the scale ladder is a series of collapse-return cycles*, each generating the degrees of freedom that the next scale requires. The confinement cliff does not merely precede nuclear binding — it *creates the conditions for it*: the dead channels of the proton are exactly what nuclear physics replaces. This is the axiom in action: collapse is generative. Destruction of coherence at one scale is the precondition for new coherence at the next.

Note a temporal consequence. The return rate c_1 climbs from the neutron's 0.60 to the alpha particle's 0.99 and iron-56's 0.98. A system with $c_1 = 0.99$ completes 99% of its collapse-return cycles; one with $c_1 = 0.60$ completes only 60%. The completed cycles *are* the clock ticks. A higher return rate means more ticks per unit of drift — a faster internal clock. Nuclear binding, by restoring the return channel, does not just restore coherence; it restores the system's capacity to experience time. The free neutron, losing 40% of its cycles to β-decay, lives in a slower internal time than the bound neutron inside iron-56.

At this scale, Δ is not merely a number but a dynamic diagnostic: it opens at confinement boundaries and closes through nuclear binding, while c_1 measures temporal resolution (Diagnostic Key, Sec. II A).

V. SCALE III: STARS, PLANETS, AND COSMOS

At stellar and galactic scales, the budget surface acquires geometric structure that maps onto familiar grav-

itational phenomena — not as a replacement for general relativity, but as a diagnostic lens that reveals *why the same accumulation pattern persists* at scales where gravity dominates.

At this scale the gap Δ ceases to be merely a diagnostic number and becomes a *classifier*: it determines lensing morphology (Theorem 12). And the budget function $\Gamma(\omega)$ (Diagnostic Key, Sec. II A) acquires the structure of a *potential*: its gradient plays the diagnostic role of gravity (Theorem 5), its convexity plays the role of tidal force (Theorem 6), and its cost asymmetry produces a directional arrow (Theorem 11). Where subatomic and nuclear scales reveal the two-step pattern arithmetically, cosmic scales reveal it geometrically.

A. Accumulation as Gradient

Theorem 5 (T-ST-1: Budget Gradient)**.** *The budget gradient* $g_{\mathrm{GCD}}(\omega) = \mathrm{d}\Gamma/\mathrm{d}\omega > 0$ *for all* $\omega \in (0,1)$*. Systems at higher drift experience steeper cost — the diagnostic analog of gravitational field strength.*

Remark 3. *Positivity is verified at 100 uniformly spaced points and for all 37 entities. The analytical form agrees with the numerical derivative to relative error* $< 10^{-5}$*. (4/4 sub-tests.)*

Theorem 6 (T-ST-2: Always Attractive)**.** $\mathrm{d}^2\Gamma/\mathrm{d}\omega^2 > 0$ *everywhere in* $(0,1)$*: the field is monotonically increasing with drift. There is no repulsive regime.*

Remark 4. *The analytical second derivative of* $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$ *is* $\Gamma''(\omega) = [6\omega(1-\omega+\varepsilon)^2 - 6\omega^2(1-\omega+\varepsilon) + 2\omega^3]/(1-\omega+\varepsilon)^3$*. Positivity is verified at 100 grid points and for all 37 entities. The minimum value across the grid is* $\Gamma''(0.01) > 0$*. (3/3 sub-tests.)*

Theorem 7 (T-ST-3: Cubic Onset)**.** *For small* ω*,* $\Gamma(\omega) \approx \omega^3$*. This cubic onset makes gravity the weakest interaction at low drift. The frozen exponent* $p = 3$ *is the unique integer for which* ω_{trap} *is a Cardano root of* $x^3 + x - 1 = 0$*.*

Remark 5. *At* $\omega = 0.01$*,* $\Gamma(0.01) = 10^{-6}/(0.99+\varepsilon) \approx 1.01 \times 10^{-6}$ *versus the cubic approximation* $\omega^3 = 10^{-6}$*, confirming the cubic-onset to relative error* $< 10^{-2}$*. The uniqueness of* $p = 3$ *is verified: for* $p \in \{1,2,3,4,5\}$*, only* $p = 3$ *produces an* ω_{trap} *that satisfies* $x^3 + x - 1 = 0$ *(Cardano root* $x \approx 0.6824$*). (3/3 sub-tests.)*

Theorem 8 (T-ST-5: Equivalence Principle)**.** $\mathrm{d}\Gamma/\mathrm{d}\omega$ *depends on* ω *alone, not on* $\mathbf{c}$ *or* $\mathbf{w}$*. All entities with the same* ω *experience identical gradient — a neutron star and a trauma well at the same drift sit on the same cost contour.*

Remark 6. *This follows algebraically:* $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$ *contains no* c_i *or* w_i *terms. The gradient* $\Gamma'(\omega)$ *is a function of* ω *alone. Verified by computing* Γ' *for all 37 entities and confirming that entities with the same* ω *(within* 10^{-6}*) produce identical gradients. (3/3 sub-tests.)*

B. The Stellar Landscape

TABLE IV. Stellar and planetary entities.

Entity	Cat.	F	IC	Δ	IC/F
Main-seq. star	Stel.	0.804	0.773	0.031	0.961
Red giant	Stel.	0.719	0.703	0.016	0.977
White dwarf	Stel.	0.753	0.713	0.039	0.948
Neutron star	Stel.	0.880	0.879	0.001	0.998
Black hole	Stel.	0.690	0.569	0.121	0.825
Magnetar	Stel.	0.825	0.822	0.003	0.997
Pulsar	Stel.	0.861	0.857	0.005	0.995
Rocky planet	Plan.	0.656	0.574	0.082	0.875
Gas giant	Plan.	0.706	0.673	0.033	0.953
Ice giant	Plan.	0.653	0.593	0.059	0.910
Dwarf planet	Plan.	0.585	0.467	0.118	0.799
Moon	Plan.	0.559	0.414	0.144	0.742

Neutron stars are the most coherent astrophysical objects. The neutron star achieves IC/$F = 0.998$ — its eight channels are nearly uniform, producing a gap of only $\Delta = 0.001$. The pulsar (IC/$F = 0.995$) and magnetar (IC/$F = 0.997$) are close behind. These objects have excellent cycle return, deep symmetric wells, and tight trajectory closure from their periodic emission.

The neutron star is the *culmination* of nuclear recovery. At the subatomic scale, nuclear binding raised IC/F from the proton's 0.718 to iron-56's 0.954. The neutron star pushes this to the limit: an entire star compressed to nuclear density, every nucleon participating in a single coherent structure. It is the most extreme version of the principle that new degrees of freedom restore dead channels — here, the "new degrees of freedom" are degenerate neutron matter, the densest stable state of baryonic matter. At the cognitive scale, the healthy memory occupies a structurally analogous position: the most coherent state in its category, with IC/$F = 0.982$ and $\Delta = 0.014$ — a perfect ring, like the neutron star.

The stellar lifecycle is a coherence evolution. Reading Table IV from main sequence to remnant tells a collapse-return story. The main-sequence star (IC/$F = 0.961$) burns hydrogen in thermodynamic equilibrium. As fuel depletes, the star expands into a red giant (IC/$F = 0.977$) — the IC/F actually *rises* because the bloated envelope, while larger, has more uniform channel distributions. At stellar death, the outcome depends on mass. Low-mass stars leave a white dwarf (IC/$F = 0.948$): a cooling remnant with moderate coherence, its trajectory closure maintained by stable electron degeneracy. High-mass stars collapse to a neutron star (0.998) or a black hole (0.825) — the most coherent and the least coherent astrophysical outcomes, respectively. The channel that separates them is trajectory closure: the neutron star's $c_5 = 0.95$ versus the black hole's $c_5 = 0.10$. One returns; the other does not.

The stellar lifecycle is also a temporal evolution. The return rate c_1 traces the star's relationship with time. The main-sequence star ($c_1 = 0.95$) has a fast, reliable clock: hydrogen fusion cycles complete smoothly. The red giant ($c_1 = 0.85$) begins to miss cycles as the core contracts and the envelope expands on different timescales — internal temporal coherence weakens. The pulsar ($c_1 = 0.98$) is the most temporally precise astrophysical object: its radio pulses are clocks accurate to nanoseconds, and its c_1 reflects this. But the black hole ($c_1 = 0.50$) has lost half its temporal resolution. Matter crosses the horizon but does not return; 50% of cycles end permanently. From the budget-surface perspective (Theorem 10), the black hole sits at $\omega = 0.310$, where $\Gamma'(0.310) = 0.48$ — each surviving tick costs nine times more than the neutron star's. The black hole experiences time as expensive and incomplete: few cycles, each costly. The neutron star experiences time as cheap and nearly perfect: many cycles, each affordable.

The black hole is the astrophysical PTSD. The black hole's $\mathrm{IC}/F = 0.825$ is the lowest among stellar objects. The mechanism is geometric slaughter: its cycle return rate ($c_1 = 0.50$) and trajectory closure ($c_5 = 0.10$) are critically low — matter falls in but does not return, and trajectories do not close. These are the same two channels that are dead in the photon ($c_5 \approx 0$) and in the PTSD well ($c_5 = 0.10$, $c_1 = 0.30$, Table VIII). The photon cannot close trajectories because it has no mass; the black hole cannot because it has too much.

Planets: shallow wells with graded coherence. The planetary objects span IC/F from 0.953 (gas giant) to 0.742 (Moon). The gas giant resembles a scaled-down main-sequence star — a deep, self-gravitating well with reliable convective return. The ice giant ($\mathrm{IC}/F = 0.910$) is intermediate: its weaker self-gravity produces shallower wells, but convective return remains reliable. The rocky planet (0.875) sits below, with its geological cycling (plate tectonics, hydrological cycle) providing moderate trajectory closure. The Moon ($\mathrm{IC}/F = 0.742$) is instructive: its dead gradient channel (c_3 low) and shallow well reflect a body too small for sustained internal dynamics. Its $\mathrm{IC}/F = 0.742$ is close to the proton's $\mathrm{IC}/F = 0.718$ — both are systems where internal freedom is too constrained for full coherence, one by confinement and the other by insufficient mass. The dwarf planet (0.799) sits between the Moon and the rocky planet, mapping the structural transition from gravitationally inert to dynamically active.

C. Accumulation of Depth: Mass as Iterated Return

Theorem 9 (T-ST-6: Memory Wells from Iteration)**.** *Well depth accumulates linearly:* $\mathrm{depth} = N\,|\Delta\kappa|$. *Mass, in this diagnostic framework, is accumulated* $|\kappa|$ *from iterated collapse-return cycles — the spatial accumulation mode of the pattern.*

Remark 7. *Linearity is exact:* $\mathrm{depth}(2N) = 2\,\mathrm{depth}(N)$ *to* $< 10^{-12}$*, verified over* $N \in \{1, 10, 100, 1000\}$ *for all 37 entities. The neutron star (*$|\Delta\kappa| = 0.128$*) reaches depth 1.28 in 10 cycles; the photon (*$|\Delta\kappa| = 18.42$*) reaches depth 184.2 in the same count, matching the extreme gravity of collapsed objects. Mass is not a property of stuff; it is a measure of how many times the system has collapsed and returned. (4/4 sub-tests.)*

D. Temporal Accumulation: Time and Its Arrow

Theorem 10 (T-ST-4: Time Dilation)**.** $\Gamma(0.90)/\Gamma(0.10) > 10^3$ *and* $\mathrm{d}^2\Gamma/\mathrm{d}\omega^2 > 0$ *everywhere. Near a deep well, each unit of drift costs exponentially more budget — a clock running on collapse-return cycles ticks more slowly — a diagnostic analog of general-relativistic time dilation [5, 6].*

Remark 8. $\Gamma(0.90) = 0.9^3/(0.10 + \varepsilon) = 7.29$ *and* $\Gamma(0.10) = 0.1^3/(0.90 + \varepsilon) = 1.11 \times 10^{-3}$*, giving a ratio of* $6.56 \times 10^3 > 10^3$*. At* $\omega = 0.99$*,* $\Gamma(0.99) = 0.99^3/(0.01 + \varepsilon) = 97.0$*, and at* $\omega = 0.999$*,* $\Gamma \approx 998$ *— the cost diverges as* $\omega \to 1$*. (3/3 sub-tests.)*

Theorem 11 (T-ST-8: Arrow of Time)**.** *The collapse-return cycle is asymmetric:* $\int_{0.5}^{0.99} \Gamma\,\mathrm{d}\omega \gg \int_{0.01}^{0.5} \Gamma\,\mathrm{d}\omega$*. Descent (collapse) is cheap; ascent (return) is expensive. This cost asymmetry, combined with convexity, produces the thermodynamic arrow.*

Remark 9. *Numerical integration gives* $\int_{0.5}^{0.99} \Gamma\,\mathrm{d}\omega = 6.35$ *versus* $\int_{0.01}^{0.5} \Gamma\,\mathrm{d}\omega = 0.016$*, a ratio of* $\approx 400:1$*. The ascent costs* $400\times$ *as much budget as the descent. This is a structural irreversibility: a system that drifts from* $\omega = 0.01$ *to* $\omega = 0.99$ *accumulates a budget debt that is enormously more expensive to repay than it was to incur. (3/3 sub-tests.)*

Why time behaves differently at different scales. Two mechanisms modulate time in this framework, and they are independent:

(i) Return rate (c_1): how many cycles complete. Each entity's internal clock ticks once per completed collapse-return cycle. If $c_1 = 0.98$ (quark), nearly every cycle completes and the clock runs fast. If $c_1 = 0.50$ (black hole), half the cycles fail to return and the clock runs at half speed. If $c_1 = 0.01$ (photon), the clock effectively stops: photons do not experience internal time because they almost never return. This is the budget-surface analog of the relativistic result that a photon's proper time is zero.

(ii) Budget cost ($\Gamma'(\omega)$): how expensive each tick is. Even when cycles complete, the cost per cycle varies. At the neutron star's $\omega = 0.120$, the budget gradient is $\Gamma'(0.120) = 0.051$ — each tick is cheap. At the black hole's $\omega = 0.310$, $\Gamma'(0.310) = 0.48$ — each tick costs nine times more. Near the pole ($\omega \to 1$), the cost diverges: the clock still ticks but each tick consumes the system's

entire remaining budget. This is time dilation: not a slowing of the clock mechanism, but an increase in what each tick costs.

These two effects combine multiplicatively. The effective temporal resolution of a system is proportional to $c_1/\Gamma'(\omega)$: the number of completed cycles divided by the cost per cycle. The neutron star ($c_1 = 0.92$, $\Gamma' = 0.051$) has high temporal resolution — many cheap ticks. The black hole ($c_1 = 0.50$, $\Gamma' = 0.48$) has low temporal resolution — few expensive ticks. The photon ($c_1 = 0.01$, $\Gamma' = 12.0$) is doubly penalized: it almost never returns *and* each tick is enormously costly — its temporal resolution $\mathcal{T} \approx 0.001$ is effectively zero.

The arrow of time (Theorem 11) enters because these costs are asymmetric. Falling into a well (increasing ω) consumes budget at cubic onset — slowly at first, then explosively. Climbing out (decreasing ω) requires repaying the accumulated cost against the convex gradient. The deeper the well, the greater the asymmetry. This is why the neutron star's clock ticks reliably but the black hole's clock struggles: the black hole sits deep enough on the budget surface that each return costs enormously more than the corresponding descent. The thermodynamic arrow, in this diagnostic framework, is this asymmetry.

This analysis compounds across scales. At the biological scale, the heart's $c_1 = 0.98$ produces the fastest biological clock (Table VII), and the immune response's $c_1 = 0.80$ produces a sluggish one. At the cognitive scale, the PTSD well's $c_1 = 0.30$ (Table VIII) explains why traumatic time feels frozen — the internal clock has nearly stopped, not because there is nothing happening, but because 70% of the cycles fail to return.

E. Lensing from Heterogeneity

Theorem 12 (T-ST-7: Gap-Controlled Lensing)**.** *The heterogeneity gap Δ controls lensing morphology: $\Delta < 0.02$ produces a perfect ring, $\Delta < 0.10$ a thick arc, $\Delta < 0.30$ a thin arc, and $\Delta \geq 0.30$ distorted lensing [7, 8]. The deflection angle $\delta\theta = 2|\kappa_{\text{well}}|/(b^2 + 0.1)$ increases with well depth and decreases with impact parameter b.*

Remark 10. *The four lensing classes partition all 37 entities consistently. Perfect ring ($\Delta < 0.02$): neutron star (0.001), magnetar (0.003), pulsar (0.005), galaxy cluster (0.006), grief (0.006), spiral galaxy (0.007), healthy memory (0.014). Thick arc ($0.02 \leq \Delta < 0.10$): quark (0.033), iron-56 (0.036), carbon-12 (0.073), habit loop (0.089). Thin arc ($0.10 \leq \Delta < 0.30$): PTSD (0.115), black hole (0.121), proton (0.134), electron (0.141), photon (0.245). No entity in this catalog reaches the distorted threshold. (4/4 sub-tests.)*

The neutron star ($\Delta = 0.001$) produces a perfect ring — the most symmetric lensing pattern possible, consistent with its position as the coherence peak of the nuclear recovery sequence that began with iron-56's $\Delta = 0.036$. The black hole ($\Delta = 0.121$) produces a thin arc — the asymmetry in its internal channels distorts the symmetric lensing pattern. This is the gravitational-lensing analogue of geometric slaughter: dead channels produce asymmetric gradients, which bend light into arcs rather than rings. Note that the black hole's lensing class (*thin arc*) is the same as the proton's ($\Delta = 0.134$) — and the same as PTSD's ($\Delta = 0.115$, Table VIII). The kernel does not know what these systems are made of. It only sees the gap.

Retroactive depth: every previous number has changed meaning. Return to Table II. The proton's $\Delta = 0.134$ is no longer just a diagnostic distance between two averages — it is a lensing classification (*thin arc*). The quark's $\Delta = 0.033$ is a *thick arc*. The photon's $\Delta = 0.245$ is a *thin arc* — the most asymmetric in the catalog. Every entry in Table III has acquired geometric meaning: helium-4's $\Delta = 0.064$ (thick arc) versus iron-56's $\Delta = 0.036$ (thick arc approaching perfect ring). The numbers have not changed. What has changed is the reader's capacity to *read* them: the budget surface gave the gap a geometry it did not have at the subatomic scale. This is the compounding at work — each section deepens every previous one.

The diagnostic toolkit now has three registers: (i) the arithmetic of slaughter (Δ as a number), (ii) the dynamics of recovery (slaughter/recovery cycles at phase boundaries), and (iii) the geometry of the gap (Δ as lensing classifier, Γ' as gravity, Γ'' as tidal force). Together they yield a further register: *prediction*.

F. The Cosmic Web

TABLE V. Diffuse and composite structures.

Entity	Cat.	F	IC	Δ	IC/F
Molecular cloud	Diff.	0.306	0.251	0.056	0.819
Dust lane	Diff.	0.235	0.182	0.053	0.773
Nebula	Diff.	0.354	0.305	0.048	0.863
Intergal. medium	Diff.	0.174	0.112	0.062	0.643
Spiral galaxy	Comp.	0.766	0.759	0.007	0.991
Galaxy cluster	Comp.	0.706	0.700	0.006	0.991
Binary star	Comp.	0.772	0.761	0.012	0.984
Accretion disk	Comp.	0.591	0.581	0.010	0.983
Cosmic filament	Comp.	0.509	0.490	0.019	0.963

The diffuse media sit in Collapse regime with low F and $\omega > 0.30$ — structurally incoherent substrates from which stars condense. Their role in the scale ladder parallels the confinement cliff: they are not failures but *preconditions*, their low coherence creating the gradient ($d\Gamma/d\omega > 0$) that drives gravitational condensation.

The *nebula* (IC/$F = 0.863$, $F = 0.354$) is the most coherent diffuse entity, with moderate trajectory closure

($c_5 = 0.35$) and enough internal structure — ionization fronts, shock boundaries — to sustain local collapse-return cycles. The *molecular cloud* (IC/F = 0.819, $F = 0.306$) sits lower: it has enough structure to fragment into clumps ($c_5 = 0.30$) but not enough to stabilize — collapse is inevitable, and from that collapse, stars are born. The *dust lane* (IC/$F = 0.773$, $F = 0.235$) is dimmer still: its near-zero gradient ($c_3 = 0.03$) reflects the absence of internal dynamics; it absorbs and obscures but does not actively cycle. The *intergalactic medium* (IC/$F = 0.643$, $F = 0.174$) is the most incoherent entity in the astrophysical catalog. Its channels are uniformly depleted ($c_3 = 0.01$, $c_6 = 0.05$) — the residuum of cosmic evolution, the substrate that remains after all condensation has occurred. The four diffuse entities form a coherence gradient: nebula > cloud > dust > IGM, tracing the structural pathway from active collapse to inert residue.

The composite structures tell the opposite story — assembly restores coherence. The *spiral galaxy* (IC/F = 0.991, $\Delta = 0.007$) achieves near-perfect uniformity from billions of stellar oscillators: its channel profile is remarkably flat ($c_0 = 0.82$, $c_1 = 0.88$, $c_5 = 0.80$, $c_7 = 0.78$), with no single channel below 0.55. The *galaxy cluster* (IC/F = 0.991, $\Delta = 0.006$) matches it through a different mechanism: where the spiral galaxy's uniformity comes from rotational averaging, the cluster's comes from statistical depth — hundreds of galaxies averaging their channel profiles into the narrowest gap in the cosmic catalog. The *binary star* (IC/$F = 0.984$, $\Delta = 0.012$) is a two-body composite whose trajectory closure ($c_5 = 0.90$) and return rate ($c_1 = 0.92$) reflect mutual orbital periodicity — the simplest case of composite coherence exceeding individual components. The *accretion disk* (IC/$F = 0.983$, $\Delta = 0.010$) achieves comparable coherence through a different route: its high gradient ($c_3 = 0.80$) and deep well ($c_2 = 0.70$) compensate for a lower return rate ($c_1 = 0.55$), reflecting matter that spirals inward reliably but does not return. It is a repetitive orbital pattern with reliable but shallow return. The *cosmic filament* (IC/$F = 0.963$, $\Delta = 0.019$) is the least coherent composite, sitting at the boundary between thick arc and perfect ring. Its moderate well depth ($c_2 = 0.55$) and broad distribution ($c_3 = 0.25$) reflect the large-scale structure of the universe: coherent enough to channel matter along gravitational corridors, but too diffuse for the tight uniformity of a galaxy or cluster.

G. The Budget Surface Itself

Theorem 13 (T-ST-9: Intrinsic Flatness)**.** *The Gaussian curvature of $z(\omega, C) = \Gamma(\omega) + \alpha\, C$ vanishes identically: $K = 0$. The surface is developable — all structure comes from the embedding.*

Remark 11. *Since z is linear in C, $\partial^2 z/\partial C^2 = 0$ and $K = 0$ follows algebraically: $K = (z_{xx} z_{yy} - z_{xy}^2)/(1 + z_x^2 + z_y^2)^2$, and $z_{yy} = 0$, $z_{xy} = 0$ force $K = 0$ identically. This mirrors a key finding from the 44 structural identities: the Bernoulli manifold is flat in Fisher coordinates ($g_F(\theta) = 1$, verified to $< 10^{-16}$ [1]). Numerical verification: K evaluated on a 100×100 grid over $\omega \in [0.01, 0.99]$, $C \in [0, 1]$ returns exactly 0 at all 10,000 points. (3/3 sub-tests.)*

Table VI summarizes the complete GR–GCD correspondence.

TABLE VI. Diagnostic correspondences between general relativity and GCD budget-surface quantities (structural parallels, not derivations).

GR Concept	GCD Budget-Surface Quantity
Mass	Accumulated $\lvert\kappa\rvert$ from iterations
Curvature	Extrinsic curvature of $z(\omega, C)$
Geodesic	Least-budget path (min-Γ trajectory)
Event horizon	$\omega \to 1$ pole ($\Gamma \to \infty$)
Graviton	Minimum $\lvert\Delta\kappa\rvert$ deposit
Time dilation	τ_R^* increases near wells
Tidal force	$\mathrm{d}^2\Gamma/\mathrm{d}\omega^2$
Einstein ring	Closed deflection orbit
Lensing shape	Controlled by $\Delta = F - \mathrm{IC}$

As $\omega \to 1$, the denominator $1 - \omega + \varepsilon \to \varepsilon$ and $\Gamma \to 1/\varepsilon = 10^8$. This pole is the budget-surface analogue of the Schwarzschild singularity: the cost of return becomes effectively infinite. The guard band ε prevents a true divergence, analogous to Planck-scale regularization.

VI. SCALE IV: THE LIVING SYSTEM

The same kernel that classifies subatomic particles and stellar objects now turns to the living body. The Diagnostic Key (Sec. II A) predicts what the data will show: periodic oscillators with reliable return should have high IC/F (the two-step pattern: periodicity provides the degrees of freedom that close the gap); composites should exceed their components (as nuclear binding exceeds hadron coherence, Table III).

Test the prediction before seeing the data. The reader can verify that the Diagnostic Key's structural logic, applied to the encoding of Table I, predicts the hierarchy before consulting Table VII. The heart is a periodic oscillator with reliable return — should its IC/F be closer to the neutron star's 0.998 (Table IV, Diagnostic Key item 7) or the dust lane's 0.773 (Table V)? The cortex assembles from individual neurons, as a galaxy cluster assembles from individual stars — should its IC/F be higher or lower than the single neuron's? The two-step pattern predicts "closer to the neutron star" and "higher than the single neuron": periodic oscillators have high IC/F; composites exceed their components. Let us check.

TABLE VII. Biological entities.

Entity	F	IC	Δ	IC/F
Neuron	0.679	0.633	0.046	0.933
Cortical network	0.622	0.606	0.016	0.974
Heart rhythm	0.769	0.730	0.039	0.949
Immune response	0.588	0.575	0.013	0.978

Four biological systems — the neuron, the cortical network, the cardiac rhythm, and the immune response — are encoded as 8-channel trace vectors using channels that carry exactly the same operational meaning as at subatomic and cosmic scales: coherence persistence, cycle return rate, well depth, gradient strength, tidal symmetry, trajectory closure, circulation area, and heterogeneity profile.

The neuron is a collapse-return oscillator. The action potential is a prototypical collapse-return cycle: depolarization collapses the resting state, and the refractory period guarantees return. The neuron's IC/F = 0.933 is comparable to the carbon atom (0.898) and the stellar main sequence (0.961) — all three are moderately bound systems whose internal heterogeneity produces a gap $\Delta \approx 0.03$–0.07 without killing any channel outright. The weakest channel is gradient strength ($c_3 = 0.30$) — the single neuron's voltage gradient is modest — which introduces a gap of $\Delta = 0.046$. But unlike the proton or the black hole, no channel drops below 0.30: the neuron avoids geometric slaughter. This is why neurons can assemble into cortical networks that *exceed* their components (cortex IC/F = 0.974 vs. neuron 0.933) — the floor is high enough to survive composition.

The heart is the fastest biological oscillator. The cardiac sinus rhythm achieves IC/F = 0.949 and F = 0.769 — the highest fidelity among biological entities. Its trajectory closure ($c_5 = 0.95$) and return rate ($c_1 = 0.98$) rival the pulsar's ($c_5 = 0.95$, $c_1 = 0.98$). The heart is a biological pulsar: both the cardiac sinus node and the pulsar magnetosphere produce rhythmic collapse-return cycles with millisecond-to-second periods, and both achieve IC/$F > 0.94$. Its Watch regime means it is the only biological system stable enough to avoid Collapse classification. When the heart's rhythm destabilizes (arrhythmia), the structural signature is a drop in c_5 and c_1 — the same channel degradation that separates the neutron star from the black hole.

The cortical network is a composite well. Like a galaxy cluster assembling from individual galaxies, the cortex assembles from individual neurons. Its IC/F = 0.974 is *higher* than the single neuron's 0.933, reflecting the power of composite structure: the network as a whole is more uniform than its parts, just as the galaxy cluster (IC/F = 0.991) exceeds the individual stellar objects. The structural reason is the same at both scales: averaging over many oscillators smooths the channel profile, raising the geometric mean toward the arithmetic mean. The gap shrinks from the neuron's $\Delta = 0.046$ to the cortex's $\Delta = 0.016$ — the same compression that takes individual stars ($\Delta \approx 0.01$–0.04) to the galaxy cluster's $\Delta = 0.006$.

The immune response is a delayed-return cycle. The inflammatory-resolution cycle has IC/F = 0.978 but F = 0.588 — moderate overall coherence. The memory B cells enable immunological return ($c_1 = 0.80$), but the tidal symmetry ($c_4 = 0.60$) reflects the chronic imbalance between pro- and anti-inflammatory signals. The immune system is structurally in Collapse ($\omega = 0.412 > 0.30$), not because it is failing, but because its breadth of response keeps drift high. This mirrors the nebula (IC/F = 0.863, $\omega > 0.30$) — both are diffuse, reactive systems operating in Collapse regime as a functional requirement rather than a pathology.

Biological time has a tempo hierarchy. The two-mechanism framework from Sec. V D applies directly. The heart ($c_1 = 0.98$, $\Gamma' = 0.23$) produces fast, cheap temporal ticks — the highest temporal resolution among biological entities. The neuron ($c_1 = 0.90$) is slower: 10% of action potentials fail to produce a spike-response at the postsynaptic terminal, and each tick costs roughly twice as much ($\Gamma' = 0.53$). The cortical network ($c_1 = 0.82$) is slower still: network-level oscillations (alpha, gamma) emerge from averaging many neuronal clocks, and the effective temporal resolution coarsens — this is why subjective time feels smoother than neuronal time. The immune system ($c_1 = 0.80$) operates on the slowest biological timescale: its inflammatory-resolution cycles take days, and 20% of cycles fail to resolve completely. This is why immune recovery *feels* slow, while heartbeats feel instantaneous. The hierarchy mirrors what we saw at the astrophysical scale (pulsar → main sequence → red giant → black hole), now re-instantiated at the cellular level: same mechanism, different substrate, same measurable ordering.

These biological well depths, gradients, and return rates are computed by the same kernel, with the same frozen parameters, using the same identities as those applied to quarks and galaxies. The mathematics does not change at the phase boundary between physics and biology. Only the channel encoding changes — and the structure that survives collapse is measured identically.

Predictions confirmed. The heart's IC/F = 0.949 sits between the pulsar (0.995) and the neutron star (0.998): periodic oscillators cluster together across twenty orders of magnitude. The cortical network's IC/F = 0.974 exceeds the single neuron's 0.933, just as the galaxy cluster (0.991) exceeds individual stellar objects. The structural pattern predicted the data. This is not a post-hoc observation — the reader was invited to make the prediction before seeing the table. The structural prediction matched the data — the pattern generalizes independently of domain expertise. The same diagnostic intuition is powerful enough to derive a clinical diagnosis from first principles (Sec. VII).

TABLE VIII. Cognitive entities.

Entity	F	IC	Δ	IC/F
Healthy memory	0.781	0.767	0.014	0.982
Trauma (PTSD)	0.481	0.366	0.115	0.760
Habit loop	0.715	0.626	0.089	0.876
Grief cycle	0.556	0.550	0.006	0.989

VII. SCALE V: THE MEMORY WELL

We arrive at the human mind. The same 8-channel kernel (Diagnostic Key, Sec. II A) now examines the most intimate structures a human being possesses: memories, habits, and grief. If the two-step pattern (geometric slaughter at phase boundaries, recovery through new degrees of freedom) operates here, then trauma — which fragments specific channels while leaving others vivid — should produce the same diagnostic signature as physical confinement.

Derive the signature before seeing the data. The Diagnostic Key provides everything needed. Construct the PTSD prediction:

1. PTSD fragments specific channels while leaving others vivid. This is heterogeneous — predict $\Delta > 0.10$.
2. Avoidance prevents narrative completion — predict trajectory closure $c_5 \approx 0.1$.
3. Intrusion-numbing asymmetry violates tidal symmetry — predict $c_4 < 0.25$.
4. The gap size places it in thin-arc lensing (Theorem 12).
5. The structural parallel is to confined quarks inside hadrons (the proton's IC/$F = 0.718$, Table II): predict IC/$F \approx 0.72$–0.76.

Compare this derivation against Table VIII.

A. Healthy Memory: The Perfect Ring

A healthy memory has $F = 0.781$, IC $= 0.767$, and $\Delta = 0.014$. All eight channels are active: high coherence persistence ($c_0 = 0.85$), excellent recall return ($c_1 = 0.92$), moderate well depth ($c_2 = 0.65$), balanced gradient ($c_3 = 0.55$), symmetric tidal forces ($c_4 = 0.90$), tight trajectory closure ($c_5 = 0.88$), moderate circulation ($c_6 = 0.60$), and uniform channel profile ($c_7 = 0.90$). The lensing classification is *perfect ring*: the gap is too small to distort the well's symmetry.

In the scale ladder, the healthy memory sits between the stellar main sequence (IC/$F = 0.961$) and the gas giant (IC/$F = 0.953$), occupying a structurally familiar position — a deep, symmetric well with reliable return. But its closest structural analog is the neutron star: both achieve $\Delta < 0.015$ (*perfect ring*), both have near-uniform channel profiles, and both sit atop the coherence peak of their respective recovery sequences. The neutron star is the culmination of nuclear recovery from confinement; the healthy memory is the culmination of cognitive integration from neural substrate. In both cases, the coherence was built through iterated collapse-return cycles across a phase boundary — nuclear binding across the hadron-nucleus boundary, and synaptic consolidation across the neuron-network boundary. The regime is WATCH: not perfectly stable (some drift from perfect coherence), but far from collapse.

B. PTSD: The Confinement Cliff of the Mind

Now compare:

$$\left.\frac{\mathrm{IC}}{F}\right|_{\text{healthy}} = 0.982 \quad \longrightarrow \quad \left.\frac{\mathrm{IC}}{F}\right|_{\text{PTSD}} = 0.760. \tag{11}$$

This drop (-22.6%) is structurally identical to the confinement cliff:

$$\left.\frac{\mathrm{IC}}{F}\right|_{\text{quark}} = 0.957 \quad \longrightarrow \quad \left.\frac{\mathrm{IC}}{F}\right|_{\text{proton}} = 0.718. \tag{12}$$

The mechanism is the same. The PTSD trace vector is:

```
(0.75, 0.30, 0.80, 0.70, 0.20, 0.10, 0.85,
                    0.15)
```

Four channels are critically low:

- $c_1 = 0.30$: **cycle return rate** — avoidance behavior means the memory is entered but not resolved. The mind falls into the well but does not return.
- $c_4 = 0.20$: **tidal symmetry** — hyperarousal and dissociation create a violently asymmetric well. The pull is strong in one direction (intrusion) and absent in the other (numbing).
- $c_5 = 0.10$: **trajectory closure** — the re-experiencing cycle does not close. The patient relives the trauma but cannot complete the narrative arc. This is the dead channel that kills IC.
- $c_7 = 0.15$: **heterogeneity profile** — extreme channel unevenness. Some aspects of the memory are vivid ($c_2 = 0.80$, $c_6 = 0.85$); others are fragmentary.

The parallels to the proton are exact (Table IX).

The proton's quarks cannot close trajectories because confinement prevents free propagation. The PTSD patient's memories cannot close trajectories because avoidance prevents narrative completion. In both cases, the dead trajectory channel destroys the geometric mean while the arithmetic mean (overall fidelity) remains moderate [9, 10].

TABLE IX. Structural parallel: confinement and PTSD. The same geometric slaughter mechanism operates at both scales.

Feature	Proton	PTSD
IC/F	0.718	0.760
Dead channels	trajectory, het.	trajectory, het.
Δ	0.134	0.115
Regime	Collapse	Collapse
Lensing	thin arc	thin arc
What is confined	quarks	traumatic content
What cannot close	quark orbits	narrative arc

Remark 12 (Encoding Sensitivity of the Parallel)**.** *The proton–PTSD parallel (Table IX) depends on two encoding decisions: (i) that "trajectory closure" (c_5) is a meaningful channel for both quark confinement and narrative completion, and (ii) that both receive low values ($c_5 = 0.15$ and $c_5 = 0.10$ respectively). If c_5 were defined differently — for example, as spatial volume closure for quarks but temporal narrative closure for memories — the mapping would be a* conceptual analogy imposed by the encoder*, not a structural discovery by the kernel. Similarly, if the PTSD trace vector were encoded with $c_5 = 0.50$ (moderate narrative closure rather than severe), the* IC/F *drop would be* $0.982 \to 0.873$ *— a mild depression rather than a confinement cliff. The parallel is* encoding-contingent*: it is precise and reproducible given the encoding in Table I, but alternative encodings of the same clinical phenomena could produce different structural classifications. What this framework offers is a* controlled comparison*: once the encoding is fixed, the structural diagnostics are determined, and the question becomes whether the same encoding scheme, applied consistently across domains, produces parallels that domain experts recognize as meaningful.*

The lensing classification confirms the analogy: both the proton and PTSD are classified as *thin arc* — moderate asymmetry distorting the well gradient into a one-sided pull. A healthy memory, like a neutron star, is a *perfect ring*.

PTSD freezes time. The two-mechanism framework explains why. With $c_1 = 0.30$, only 30% of collapse-return cycles complete — 70% of temporal ticks are lost. Simultaneously, $\Gamma'(0.519) = 2.28$ makes each surviving tick 11× more expensive than a healthy memory's. The effective temporal resolution $c_1/\Gamma' = 0.30/2.28 = 0.13$ is the lowest of any entity in this paper — worse even than the black hole ($c_1/\Gamma' = 0.50/0.48 = 1.04$), because the trauma well combines low return rate with a steep budget gradient. This is the structural substrate of what clinicians call *temporal disintegration*: PTSD patients report that the traumatic event feels as though it is happening "right now," that time between intrusions collapses, that minutes in a trigger state feel like hours. In the diagnostic framework, this corresponds to an accurate reading of the system's temporal structure: time, as measured by $\mathcal{T} = c_1/\Gamma'$, moves differently in a fragmented well.

Clinical consequence arising from pure geometry. The therapeutic protocol writes itself from the mathematics: restore c_5 (trajectory closure) through narrative completion — prolonged exposure therapy; restore c_4 (tidal symmetry) through grounding techniques that equilibrate intrusion and numbing; restore c_1 (return rate) through graduated re-entry and desensitization. The *order* matters: the budget gradient $\Gamma'(\omega)$ at the trauma's $\omega = 0.519$ is 11× stronger than at the healthy memory's $\omega = 0.219$, so recovery must proceed by small steps down the budget surface, closing channels one at a time rather than all at once — exactly as nuclear binding replaces dead hadron channels incrementally through shell filling and pairing.

The diagnostic toolkit has accumulated its final register: arithmetic (the gap), dynamics (slaughter and recovery), geometry (lensing and gravity), prediction (biology confirmed structural expectations), and now *consequence* — the mathematics tells a clinician which channels are dead and in what order they should be restored. Each register enriches all the others.

The black hole parallel. The black hole (IC/F = 0.825) has dead trajectory closure ($c_5 = 0.10$) and reduced return rate ($c_1 = 0.50$) — the same two channels that are near-dead in PTSD ($c_5 = 0.10$, $c_1 = 0.30$). The clinical observation that PTSD creates an "inescapable pull" toward re-experiencing has a diagnostic counterpart in this framework: the budget gradient $\mathrm{d}\Gamma/\mathrm{d}\omega$ at the trauma well's $\omega = 0.519$ produces a pull of $\Gamma'(0.519) = 2.28$, while the healthy memory at $\omega = 0.219$ experiences $\Gamma'(0.219) = 0.20$ — an 11× stronger gravitational pull.

As $\omega \to 1$, the cost of return diverges: $\Gamma \to 1/\varepsilon = 10^8$. This is the event horizon of the budget surface. PTSD is a system at $\omega = 0.519$, well within the Collapse regime ($\omega \geq 0.30$), experiencing significant but not infinite cost of return. The therapeutic implication is structural: recovery requires restoring the dead channels — closing the trajectory (c_5), balancing the tidal symmetry (c_4), improving the return rate (c_1) — exactly as nuclear binding restores the channels that confinement killed.

C. The Habit Loop: A Shallow Symmetric Well

The habit loop ($F = 0.715$, IC = 0.626, $\Delta = 0.089$) is the structural opposite of PTSD. Its trajectory closure ($c_5 = 0.92$) and return rate ($c_1 = 0.95$) are the highest among cognitive entities. The well is shallow (moderate depth, low gradient) and symmetric (high tidal symmetry). The regime is Watch: the habit is stable enough to avoid collapse, maintained by reliable cycle return.

The habit loop's closest astrophysical analog is the rocky planet (IC/F = 0.875, $\Delta = 0.082$): both are shallow wells maintained by repetitive orbital dynamics rather than deep binding. The planet orbits its star; the habit loop orbits its cue-routine-reward cycle. Neither

has the depth of the neutron star or the healthy memory, but both have the trajectory closure that the black hole and PTSD lack. The accretion disk ($\mathrm{IC}/F = 0.983$) offers another parallel: a repetitive structure where material circulates reliably, but the well is defined by the orbit rather than by internal binding.

Where PTSD is a black hole — deep, asymmetric, hard to escape — the habit loop is a gentle orbit, a well-worn track in the budget surface. Temporally, the habit loop is the cognitive inverse of PTSD: with $c_1 = 0.95$ and $\Gamma'(0.285) = 0.39$, its effective temporal resolution $c_1/\Gamma' = 2.46$ is the highest among cognitive entities. Time in a habit loop moves *fast* — cycles complete reliably and cheaply, producing the subjective experience of time vanishing: hours pass in routine without being noticed. This is the cognitive analog of the pulsar ($c_1/\Gamma' \approx 14$), where temporal ticks are so numerous and so cheap that each individual tick carries almost no experiential weight. The clinical correlate is automatic behavior: habits feel timeless precisely because their temporal resolution is so high that individual moments blur together. The lensing classification is *thick arc*: some heterogeneity ($\Delta = 0.089$) from the depth and gradient channels, but no channel is dead enough to produce geometric slaughter. This is why habits are easy to break compared to trauma: the budget gradient at the habit's $\omega = 0.285$ is $\Gamma'(0.285) = 0.39$, six times weaker than at the PTSD well's $\omega = 0.519$. The pull is real but escapable.

D. Grief: Uniform Depression

Grief ($F = 0.556$, $\mathrm{IC} = 0.550$, $\Delta = 0.006$) is structurally distinct from PTSD. Its $\mathrm{IC}/F = 0.989$ is the *highest* among cognitive entities — the channels are uniformly depressed rather than selectively killed. The lensing classification is *perfect ring*: the well is symmetric, with no dominant dead channel.

The grief cycle's closest astrophysical analog is not the black hole — it is the dust lane ($\mathrm{IC}/F = 0.773$, $\Delta = 0.053$) or the molecular cloud ($\mathrm{IC}/F = 0.819$, $\Delta = 0.056$). Like diffuse astrophysical media, grief is characterized by uniformly low coherence rather than selective channel death. Everything is reduced; nothing is shattered. The dust lane is dim across all wavelengths; the grieving mind is diminished across all functions. But crucially, the IC/F remains high because the reduction is symmetric — the geometric mean tracks the arithmetic mean closely. This is why, unlike PTSD, grief does not produce geometric slaughter: there is no single dead channel dragging the geometric mean to the floor.

This captures a clinical truth. In grief, everything hurts — all channels are moderately reduced. In PTSD, specific channels are shattered while others remain vivid. The kernel distinguishes these two states precisely: grief has low F (low overall coherence) but high IC/F (uniform channels); PTSD has moderate F but low IC/F (fragmented channels). The heterogeneity gap Δ tells the difference: $\Delta_{\mathrm{grief}} = 0.006$, $\Delta_{\mathrm{PTSD}} = 0.115$ — a $19\times$ ratio. Grief's IC/F (0.989) exceeds the healthy memory's (0.982) — a structural paradox that resolves once one recognizes that IC/F measures channel *uniformity*, not well-being: grief is uniformly diminished, and uniformity is what the geometric mean rewards.

Temporally, grief occupies the middle ground. With $c_1 = 0.55$ and $\Gamma'(0.444) = 1.34$, the effective temporal resolution is $c_1/\Gamma' = 0.41$ — slower than the healthy memory (4.58) and the habit loop (2.46), but faster than PTSD (0.13). Grief does not freeze time; it *slows* it. The grieving person experiences days that feel like weeks, not because cycles fail to complete (as in PTSD), but because each cycle costs more than it should. The uniform channel depression means that every domain of experience — work, rest, social interaction — is equally sluggish. There is no fragmentation, no dead channel, no frozen moment; there is only a pervasive temporal drag, as though the entire system is wading through the budget surface at a higher altitude than normal.

The therapeutic implication follows from the structure. PTSD recovery requires restoring specific dead channels (trajectory closure, tidal symmetry) — targeted intervention, like nuclear binding targeting specific dead hadron channels. Grief recovery requires raising all channels simultaneously — a general restoration of coherence, more akin to how a molecular cloud condenses into a star: the whole substrate must reach sufficient density before structure can nucleate.

VIII. THE UNIFICATION

Theorem 14 (T-ST-10: Cross-Scale Universality)**.** *All 37 entities in all 8 categories obey the three Tier-1 kernel identities:*

1. $F + \omega = 1$ *to* $< 10^{-12}$ *for all entities.*
2. $\mathrm{IC} \leq F$ *for all entities.*
3. $\mathrm{IC} = e^{\kappa}$ *to* $< 10^{-10}$ *for all entities.*

The trauma well has lower IC *than the healthy memory* ($\mathrm{IC}_{PTSD} = 0.366 < \mathrm{IC}_{healthy} = 0.767$)*. All 8 categories are represented and non-empty.*

Remark 13. *The universality is not a claim that quarks and memories "are the same." It is the observation that a consistent encoding scheme, when processed through the kernel, produces the same diagnostic pattern at every scale: geometric slaughter at phase boundaries followed by recovery through new degrees of freedom. The three algebraic identities are structural constraints that hold for* any *input in* $[0,1]^n$ *— they are mathematical, not empirical. What is empirical is the encoding: the channel values assigned from domain-specific observables. The significant finding is not that the identities hold (they*

must) but that the diagnostic values *(F,* IC*, Δ, $\mathcal{T}$) cluster into the same structural types (geometric slaughter with* $\mathrm{IC}/F \approx 0.7$*, perfect rings with* $\Delta < 0.02$*, cost-frozen temporal resolution) across systems separated by twenty orders of magnitude. Whether this clustering reflects a genuine structural motif of nature or a systematic feature of how humans encode oscillatory systems remains an open question — but the pattern, once exhibited, is precise enough to be falsified (Remark 2). (6/6 sub-tests.)*

The complete scale ladder is summarized in Table X.

At every phase boundary — quarks to hadrons, hadrons to nuclei, diffuse gas to stars, substrate to organism, neurons to organized memory — the same two-step process operates:

1. **Geometric slaughter at the boundary.** Dead channels in the trace vector destroy the geometric mean IC while the arithmetic mean F survives. The heterogeneity gap Δ opens.

2. **Recovery through new degrees of freedom.** New structure (nuclear binding, stellar fusion, synaptic integration, narrative completion) replaces dead channels, closing the gap and restoring IC/F toward unity.

This pattern is the content of Axiom-0: *collapse is generative*. The confinement cliff is not a catastrophe; it is the precondition for nuclear binding. The PTSD well is not a permanent state; it is the precondition for therapeutic recovery. *Only what returns is real.*

Remark 14 (Entity Selection and Coverage)**.** *The 37 entities were selected to span the scale ladder from* 10^{-15} *m to macroscopic scales, with at least three entities per category. This is a* curated catalog*, not a representative sample: the entities were chosen because they are well-studied systems with known physical properties that could be encoded in the 8-channel schema. Systems that would challenge the pattern are instructive to consider. Glasses and amorphous solids lack periodic collapse-return cycles, making the cycle_return_rate (c_1) ambiguous. Turbulent flows have well-defined energy cascades but no obvious trajectory closure. Quantum decoherence events are irreversible by definition, yielding* $c_1 \approx 0$ *and* $\mathcal{T} \approx 0$ *— these would occupy the same structural position as the photon (effectively frozen time) rather than contradicting the pattern. The two-step pattern (geometric slaughter followed by recovery) has not been tested on systems where the recovery step fails permanently — for example, radioactive decay chains that terminate at stable isotopes without restoring dead channels. Extending the catalog to include such boundary cases is necessary before the structural claims can be considered robust.*

Compounding as structural enrichment. The fact that understanding deepens through this paper is not a rhetorical device — it is a structural feature of single-axiom systems. Because every identity, every theorem, and every classification derives from the same source (Axiom-0 and the budget Γ), each new derivation enriches every previous one. The gap Δ is richer as a lensing classifier (Section V) than as a bare number (Section III), and the clinical protocol (Section VII) is richer after one has seen the recovery mechanism at nuclear scales.

But each section is also independently coherent: the Diagnostic Key (Sec. II A) provides the fixed toolkit, and each section makes its own case against it. Just as the kernel reads any domain through the same invariants, each section can be read through the same Diagnostic Key. The compounding is real — each section deepens all others — but it is an enrichment, not a prerequisite. This is the *geometry of compounding*: in a system with a single axiom, every derivation strengthens the whole, yet each derivation stands on the axiom alone.

IX. REPRODUCIBILITY

Every theorem in this paper is machine-verified. The reference implementation [11] includes the full entity catalog (37 entities, 8 categories), the kernel computation, and the theorem-proving harness. To reproduce all results:

```
pip install -e ".[all]"
python -c "
from closures.spacetime_memory \
  .spacetime_theorems import prove_all_theorems
results = prove_all_theorems()
for r in results:
    print(f'{r.name}: {r.verdict}'
          f' ({r.n_passed}/{r.n_tests})')
"
```

Output: 10/10 PROVEN, 37/37 sub-tests, 0 failures. The full test suite (`pytest tests/test_253_spacetime_memory_theorems.py -v`) runs 129 tests in < 300 ms with zero failures.

The theorem suite is structured as follows. Each theorem object specifies its name, statement, and a list of sub-tests. Sub-tests are predicate functions that return `True` or `False`; a theorem is PROVEN only if all sub-tests pass. The sub-test counts reported in the Remarks throughout this paper (e.g., "4/4 sub-tests" for Theorem 5) correspond exactly to the programmatic sub-tests in the harness. No theorem is accepted on the basis of spot-checks or manual verification.

The entity catalog is defined in `spacetime_kernel.py`, where each entity's 8-channel trace vector is frozen as a Python tuple. Modifying any channel value and rerunning the theorem suite will produce different F, IC, and Δ values — and may cause theorems to fail, demonstrating the encoding-sensitivity discussed in Remark 2. This is by design: the theorems test the *current* encoding, not a universal truth.

X. CONCLUSIONS

We have traced the geometry of what returns from quarks to memory wells, using a single axiom, a single kernel, and a single budget surface. Across 37 entities and 8 categories spanning twenty orders of magnitude, one structural pattern recurs: *structure accumulates through iterated collapse-return cycles*, and the kernel measures how that accumulation proceeds. The pattern has three modes:

TABLE X. The scale ladder: geometric slaughter and recovery across twenty orders of magnitude. $\langle \mathrm{IC}/F \rangle$ is the category mean.

Scale	Category	Entities	$\langle F \rangle$	$\langle \mathrm{IC}/F \rangle$	Structural Signature
10^{-15} m	Subatomic	5	0.509	0.656	Confinement cliff: quark 0.957 → proton 0.718
10^{-14} m	Nuclear/Atomic	3	0.747	0.923	Recovery: binding restores dead channels
10^{9}–10^{4} m	Stellar	7	0.795	0.957	Deep symmetric wells; neutron star 0.998
10^{6}–10^{8} m	Planetary	5	0.632	0.856	Shallower wells; Moon lowest at 0.742
10^{16}–10^{23} m	Diffuse	4	0.267	0.775	Low F, high ω — precursors to structure
10^{20}–10^{25} m	Composite	5	0.669	0.982	Assembled wells — highest category IC/F
10^{-6}–10^{-3} m	Biological	4	0.665	0.959	Living oscillators: heart rivals the pulsar
—	Cognitive	4	0.633	0.902	Memory wells: PTSD = confinement of the mind

- **Space accumulates as iterated well depth.** Mass, in this diagnostic framework, is accumulated $|\kappa|$ from repeated collapse-return cycles (Theorem 9). A proton's well depth is the accumulated confinement of three quarks. A neutron star's depth is the accumulated compression of 10^{57} nucleons. A trauma well's depth is the accumulated weight of repeated re-experiencing. The budget gradient ($\mathrm{d}\Gamma/\mathrm{d}\omega > 0$, Theorem 5) ensures that deeper wells cost more to escape — the same diagnostic structure that general relativity calls gravitational binding.

- **Time accumulates as affordable ticks.** A system's internal time runs at the rate at which it can complete and afford collapse-return cycles ($\mathcal{T} = c_1/\Gamma'$, Definition 4). The neutron star ticks cheaply and reliably ($\mathcal{T} \approx 18$); the PTSD well ticks rarely and expensively ($\mathcal{T} = 0.13$); the photon barely ticks at all ($\mathcal{T} \approx 0$). The cost asymmetry (descent cheap, ascent expensive, Theorem 11) produces a directional arrow: time runs forward because return costs more than collapse.

- **Memory accumulates as well geometry.** The shape of a well — its channel profile, its gap Δ, its lensing classification — is the accumulated record of how collapse-return cycles have proceeded. A healthy memory is a perfect ring ($\Delta = 0.014$): all channels uniformly reinforced by repeated successful recall. PTSD is a thin arc ($\Delta = 0.115$): selective channels dead from avoidance, others vivid from intrusion. Grief is a uniform depression ($\Delta = 0.006$): all channels lowered, none shattered. The well geometry *is* the memory — not a representation of it, but the structural residue of every cycle that built it.

The three modes are not independent. Spatial depth, temporal resolution, and well geometry are three readings of the same underlying process: iterated collapse-return. Mass is the depth the iterations have carved. Time is the rate at which iterations proceed. Memory is the shape the iterations have left. At every phase boundary — quarks to hadrons, diffuse gas to stars, neurons to organized memory — the two-step pattern operates: geometric slaughter destroys coherence at the boundary, and new degrees of freedom restore it on the other side.

What this paper has shown. The introduction made five claims — three structural, two mathematical. We close the loop:

Structural claims (encoding-dependent; falsifiable given a fixed encoding scheme):

1. **Space accumulates as iterated** $|\kappa|$ (Theorem 9). Linearity verified: $\mathrm{depth}(2N) = 2\,\mathrm{depth}(N)$ to $< 10^{-12}$ for all entities. *Falsified by*: an entity, under the same encoding, whose well depth does not grow linearly with iteration count. (4/4 sub-tests.)

2. **Time emerges from return cost** (Definition 4, Theorems 10, 11). $\mathcal{T} = c_1/\Gamma'$ ranks all 37 entities from photon (≈ 0) to neutron star (≈ 18). Cost ratio: $\Gamma(0.90)/\Gamma(0.10) = 6.56 \times 10^3$. *Falsified by*: an entity whose independently established temporal ordering contradicts $\mathcal{T}$ (Remark 1). (3/3 + 3/3 sub-tests.)

3. **Memory accumulates as well geometry** (Definition 3, Theorems 9, 14). Confinement cliff: IC/F drops $0.957 \to 0.718$ across color confinement; trauma cliff: $0.982 \to 0.760$ across narrative closure. Same dead channels (c_5, c_7), same mechanism (Remark 12). *Falsified by*: a phase boundary, under the same encoding, where the IC/F drop is not explained by dead channels.

Mathematical claims (hold for any valid input; verify implementation correctness):

4. **Budget gradient structures the manifold** (Theorem 5). $\mathrm{d}\Gamma/\mathrm{d}\omega > 0$ at 100 grid points and all 37 positions; minimum $\Gamma'(0.01) = 3.0 \times 10^{-4}$. This is a mathematical property of Γ with $p = 3$. (4/4 sub-tests.)

5. **Structural consistency across all scales** (Theorem 14). $F + \omega = 1$ to $< 10^{-12}$; $\mathrm{IC} \leq F$ with zero

violations; IC $= e^{\kappa}$ to $< 10^{-10}$. All 8 categories represented. These identities hold for any input in $[0,1]^n$; they verify the implementation, not the encoding. (6/6 sub-tests.)

The structural parallel. The parallel between the proton and PTSD (Table IX) is the paper's most striking finding: both exhibit IC/$F \approx 0.7$, both have dead trajectory closure and heterogeneity channels, both are classified as Collapse with thin-arc lensing. The proton's quarks are confined because they cannot propagate freely; the PTSD patient's trauma is confined because the narrative cannot close. The diagnostic mathematics is the same. The recovery pathway is the same: restore dead channels, close the trajectory, reduce the gap.

This parallel depends on the encoding (Remark 2): the channel values were assigned from domain-specific observables, and different assignments would produce different diagnostics. What is noteworthy is that when domain experts encode confinement physics and clinical trauma using the same 8-channel schema, the kernel reports the same structural signature. Whether this reflects a deep structural isomorphism between physical and psychological confinement, or a systematic feature of how humans conceptualize constraint, is a question the framework makes precise enough to investigate.

What the budget surface is not. No new physics is introduced. The budget surface does not replace general relativity or neuroscience. It provides a *diagnostic lens* — domain-agnostic, computationally verifiable, and continuous-integration–ready — through which phenomena separated by twenty orders of magnitude in scale become structurally visible as instances of a single accumulation pattern. The diagnostic mappings to GR concepts (Table VI) are structural correspondences, not derivations — they reveal that the budget surface has the same qualitative features (monotonic gradient, convexity, pole at $\omega \to 1$) as a gravitational potential, without claiming to produce Einstein's field equations.

The triple accumulation. The reader who began this paper with a question — "What does a quark have in common with a traumatic memory?" — now holds the answer as a pattern: both are systems where collapse-return cycles have accumulated into a well whose geometry the kernel can measure. The quark's well is spatial: accumulated $|\kappa|$ from color confinement. The star's well is temporal: its internal clock ticks at a rate set by the budget gradient. The memory's well is geometric: its shape records every recall cycle that carved it. Space, time, and memory are three faces of the same accumulation process — three ways that iterated collapse-return leaves a structural residue that the kernel reads as F, IC, and Δ.

That progression of understanding — from arithmetic, to dynamics, to geometry, to prediction, to clinical consequence — is the enrichment that linear reading adds. But the paper can also be entered at any section: the Diagnostic Key (Sec. II A) is the fixed lens through which every scale passes, and the same two-step pattern emerges regardless of entry point. The paper enacts the geometry it describes. Understanding that compounds is understanding that returns — and understanding that can be entered from any angle is understanding whose coherence is structural, not sequential.

Collapsus generativus est; solum quod redit, reale est.

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[1] C. Paulus, Universal measurement contract protocol, Manuscript (2025), contract-first measurement discipline; Tier-1 invariants reserved.
[2] C. Paulus, The physics of coherence: Recursive collapse & continuity laws (2025), pOST canon anchor; Weld-ID: W-2025-12-31-PHYS-COHERENCE.
[3] C. Paulus, Umcp casepack publication (2026), runnable CasePack publication anchor.
[4] C. Paulus, The episteme of return (2025), pRE canon anchor for UMCP/GCD.
[5] A. Einstein, Die grundlage der allgemeinen relativitätstheorie, Annalen der Physik **354**, 769 (1916).
[6] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation* (W. H. Freeman, San Francisco, 1973).
[7] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, Berlin, 1992).
[8] S. Refsdal, The gravitational lens effect, Monthly Notices of the Royal Astronomical Society **128**, 295 (1964).
[9] A. Ehlers and D. M. Clark, A cognitive model of post-traumatic stress disorder, Behaviour Research and Therapy **38**, 319 (2000).
[10] B. A. van der Kolk, *The Body Keeps the Score: Brain, Mind, and Body in the Healing of Trauma* (Viking, New York, 2014).
[11] C. Paulus, Generative-collapse-dynamics (2026), reference implementation repository (GitHub).